MEMS 5701 – Aerospace Propulsion – Tomas Collado, Eleni Kambouris, Amanda McLaughin

1. Introduction

This report presents the design and analysis of two propulsion systems: the Air-Augmented Rocket (AAR) and the Air Turbo-Rocket Ramjet (ATRR). These systems are similar in simplistic set up except for the addition of a fan in the ATRR.

2. Project Analysis

The Analysis of this project involved the application of conservation principles to define performance parameters for an AAR and ATRR. This is accompanied by trend data iterated over the inlet area of the system. These system performance diagnostics can be used in design insights going forward.

3. System Assumptions and Properties

- Gamma (γ): 1.4 for all mixtures.
- Molecular Weights: O2 = 32, H2 = 2, H2O = 18, N2 = 28.164.
- Propellant Feed Temperature: 540°R.
- Chamber Pressure: 2 × 10^5 psf.
- Rubber Engine: Area adjusts to match freestream capture.

4. Part I: Air-Augmented Rocket (AAR)

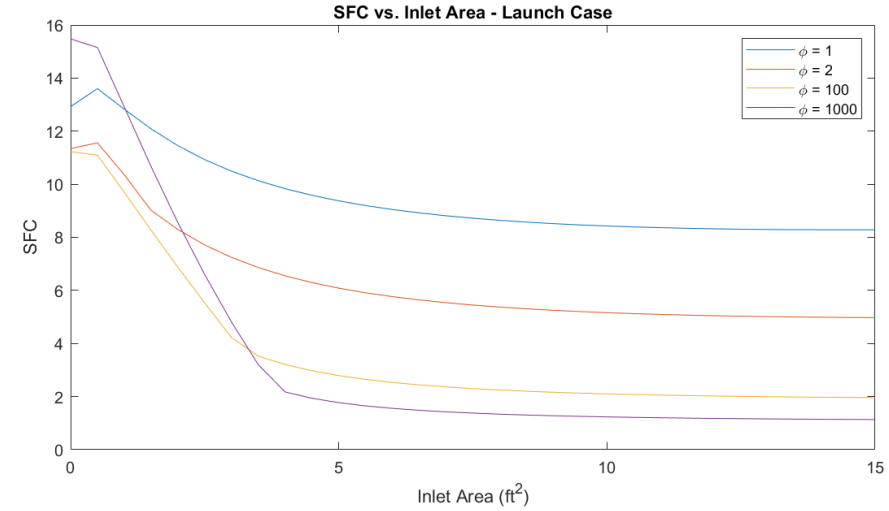
4.1 Flight Conditions

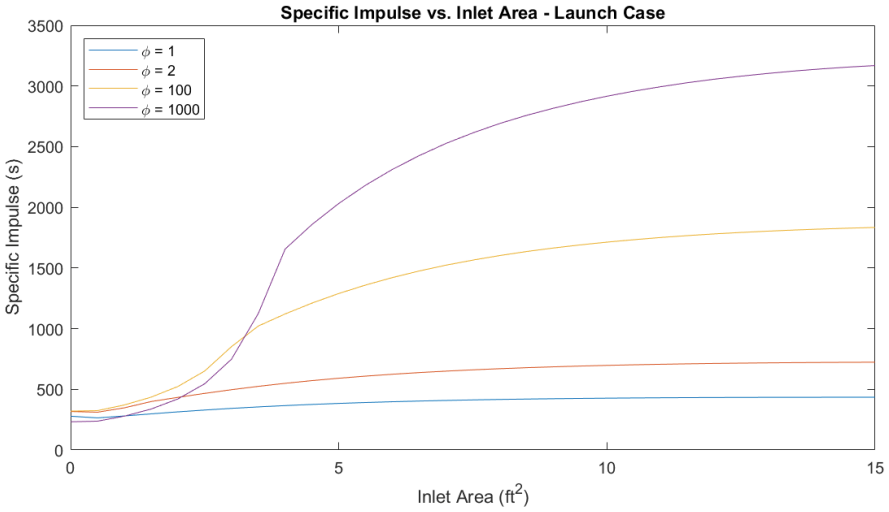
- Mach 0.8 at 30,000 ft → Required thrust: 4000 lbf
- Mach 4.0 at 60,000 ft → Required thrust: 5000 lbf

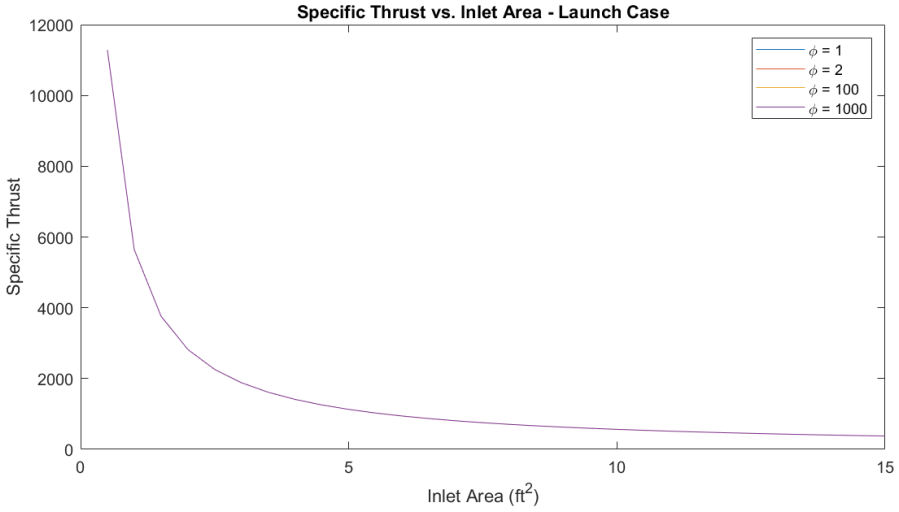
4.2 Analysis Tasks

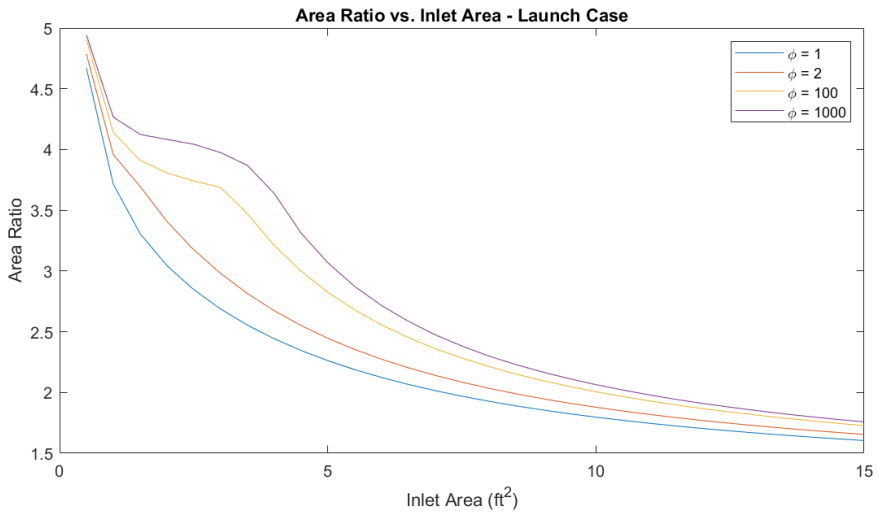
- Iterate on chamber propellant flow to match thrust.
- Vary capture area A0 from ~0 to 15 ft^2.
- Use equivalence ratios φ = 1, 2, 10, 1000.

4.3 Required Plots and Data







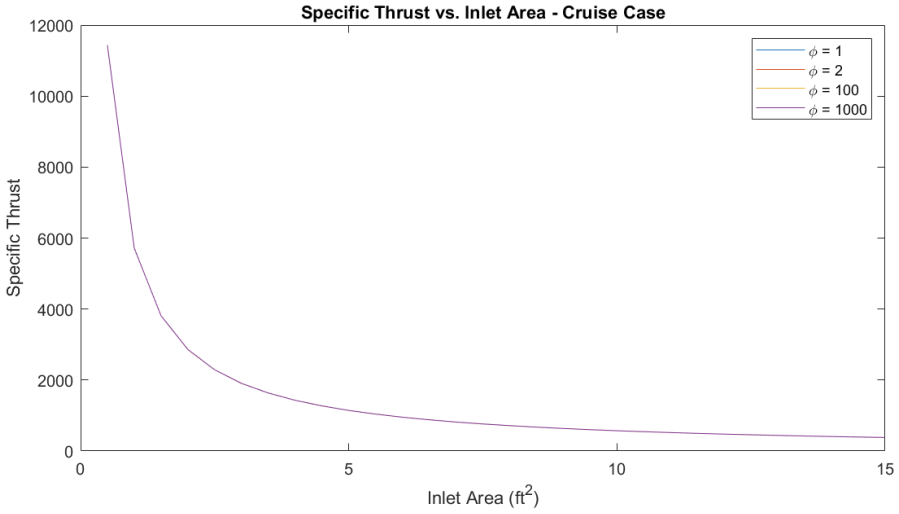


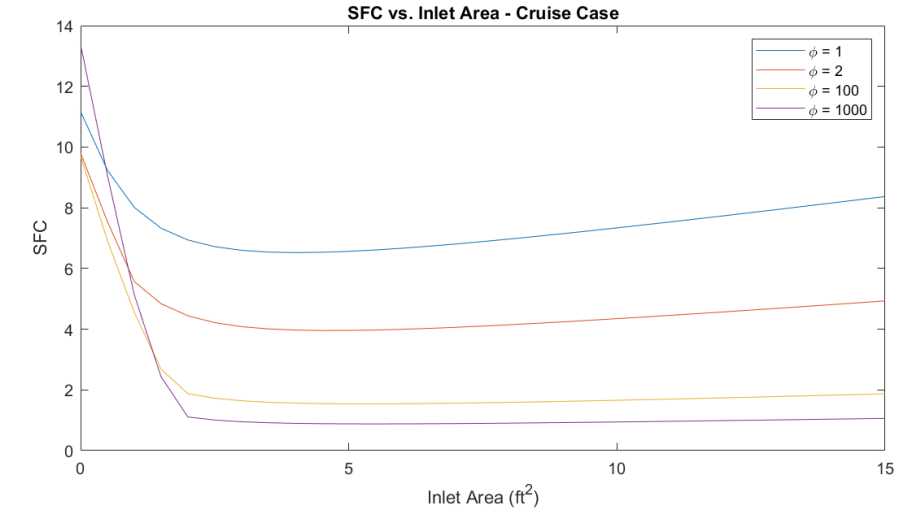
Specific fuel consumption vs. Inlet area- Specific fuel consumption decreases as the inlet area increases until reaching a plateau. High equivalence rations result in lower fuel consumption. This is expected due to the efficiency of the combustion. Mid-range inlet sizes and higher equivalence ratios are optimal for launch efficiency.

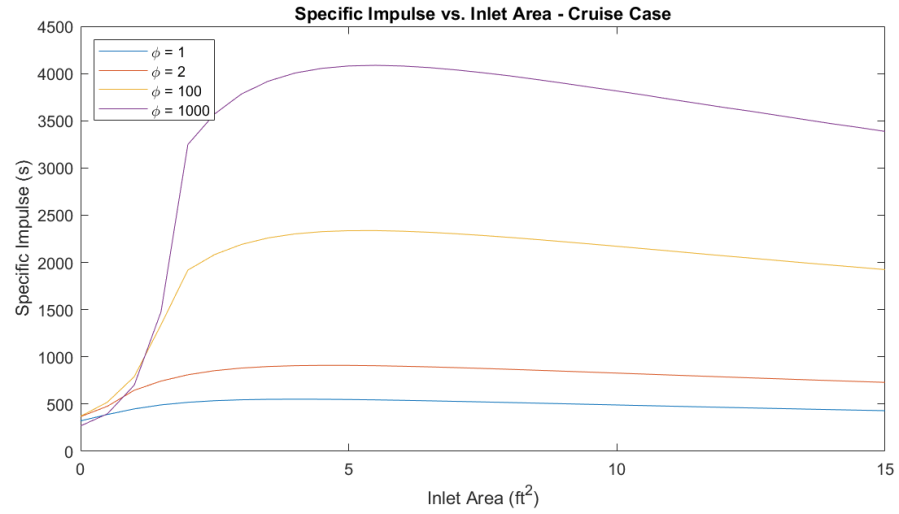
Specific impulse vs. Inlet area- The I_{sp} increases before plateauing for larger inlet areas. The I_{sp} does yield higher with an increasing equivalence ratio reaching peaks over 3000s for ϕ = 1000

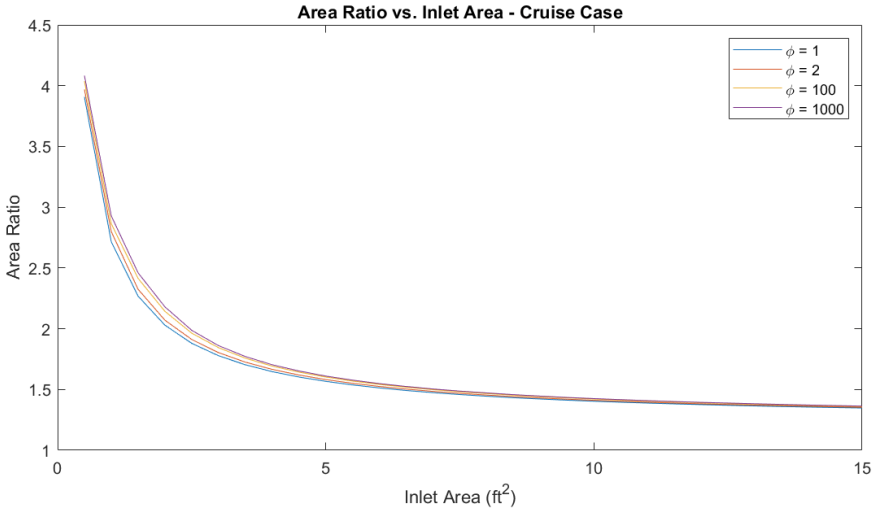
Specific Thrust vs. Inlet Area- Specific thrust is highest when the inlet area is as small as possible, especially at higher equivalence ratios. This would work well in an ideal case but may run into issues with flow constraints. Larger inlets thus improve the overall efficiency and control of the system.

Area Ratio vs. Inlet Area- Required nozzle area ratio decreases as the inlet area increases for all the integrated ratios. The larger the ratio the larger need for larger area ratios as the demand for higher expansion increases. This happens more as the inlet area decreases.


Specific Thrust vs. Inlet Area - Cruise Case
Specific Thrust
Inlet Area (ft^2)
ϕ = 1
ϕ = 2
ϕ = 100
ϕ = 1000
0
2000
4000
6000
8000
10000
12000
0
5
10
15


SFC vs. Inlet Area - Cruise Case
SFC
Inlet Area (ft^2)
ϕ = 1
ϕ = 2
ϕ = 100
ϕ = 1000
0
2
4
6
8
10
12
14
0
5
10
15


Specific Impulse vs. Inlet Area - Cruise Case
Specific Impulse (s)
Inlet Area (ft^2)
ϕ = 1
ϕ = 2
ϕ = 100
ϕ = 1000
0
500
1000
1500
2000
2500
3000
3500
4000
4500
0
5
10
15



Specific fuel consumption vs. Inlet area- The Specific fuel consumption (SFC) decreases as the inlet area increases before plateauing very early into the middle ranges of area. The equivalence ratio increases corelates with lower SFC. After the range of areas optimizes and flattens. This suggests that returns are lower with oversized outlets.

Specific impulse vs. Inlet area- The I_{sp} Increase with the inlet area quickly before hitting a peak and creating a slow decrease. The higher the equivalence ratio the higher the I_{sp} spikes and later plateaus. This suggests that at lower ratios there is limited heat release and lowers work done with expansion.

Specific Thrust vs. Inlet Area- Thrust is highest at the smallest inlet areas and decreases rapidly as the inlet enlarges. This holds to be true over all ϕ values. Smaller inlets concentrate combustion energy, but are less favorable at high speed cruise as drag penalties effect flow constraints.

Area Ratio vs. Inlet Area- The nozzle area ratio decrease with the inlet area increase. This reflect better pressure recovery and expansions at larger areas. There is little difference between equivalence ratios meaning that the performance due to area is not effected.

4.4 Discussion Prompts

- What is the maximum Isp and how does it compare to a pure rocket?

The maximum Isp is 3168 s for the launch case, and 4087 s for the cruise case. This is a very large improvement over that of a pure rocket. A hydrolox rocket tends to have Isps of approximately 400 s, so the Isp in the launch case is almost 8 times that of a pure rocket, and almost 10 times that of a pure rocket in the cruise case.

- What performance limitations appear at Mach 0.8?

The efficiency of the rocket, represented by the ISP, drops considerably at Mach 0.8 in comparison to Mach 4.0, meaning that more fuel, and thus weight, is needed to get the same range that a lighter craft traveling at Mach 4.0 would have.
- *How do the inlet, mixer, and nozzle areas behave? What changes occur when geometry is fixed (non-rubber)?*

As the inlet area increases, the nozzle area starts to decrease in size rapidly, before eventually stabilizing at a bit less than 1.5 times the inlet area. Meanwhile, the mixer area . With non-rubber geometry, the inlet, mixer, and nozzle areas would all need to be optimized for one condition, most likely the cruise condition, as most crafts spend the majority of their flight in cruise. On the other hand, this would also mean that the areas would not be optimized for the launch condition, meaning the rocket would have a lower Isp, and thus lower efficiency until it reaches its cruise altitude and speed.

Table 1: Detailed station properties for φ = 10, A0 = 4.

See Appendix, or attached Google Sheet

https://docs.google.com/spreadsheets/d/13G1dATEa26tEEdhAg7wgPHcRKmOPR5iPEEfFGixbHek/edit?usp=sharing

5. Part II: Air Turbo-Rocket Ramjet (ATRR)

5.1 Additional Assumptions

- Fan Efficiency: 95%
- Turbine Efficiency: 90%
- Shaft Efficiency: 99%
- PX = P15 and Mach X = 1.0

Include Figure 3: Schematic of the Air Turbo-Rocket Ramjet (ATRR).

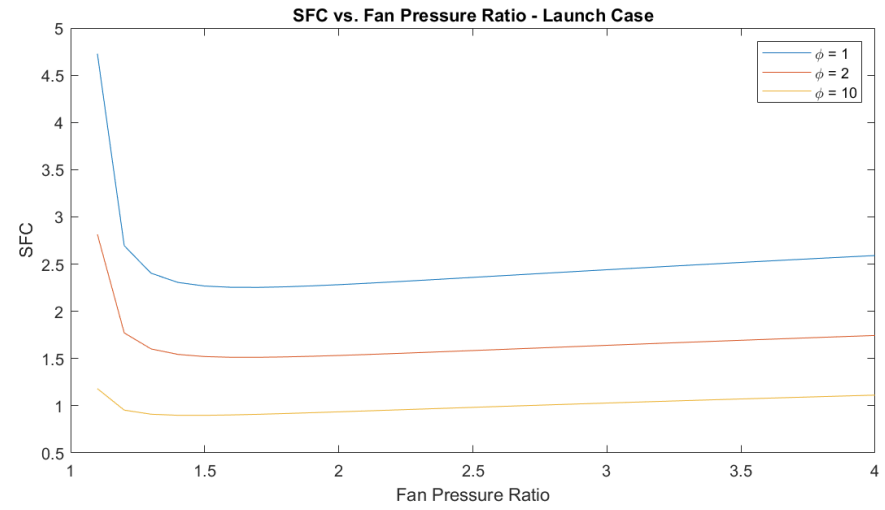
Include Figure 4: Iteration Flowchart for ATRR Analysis.

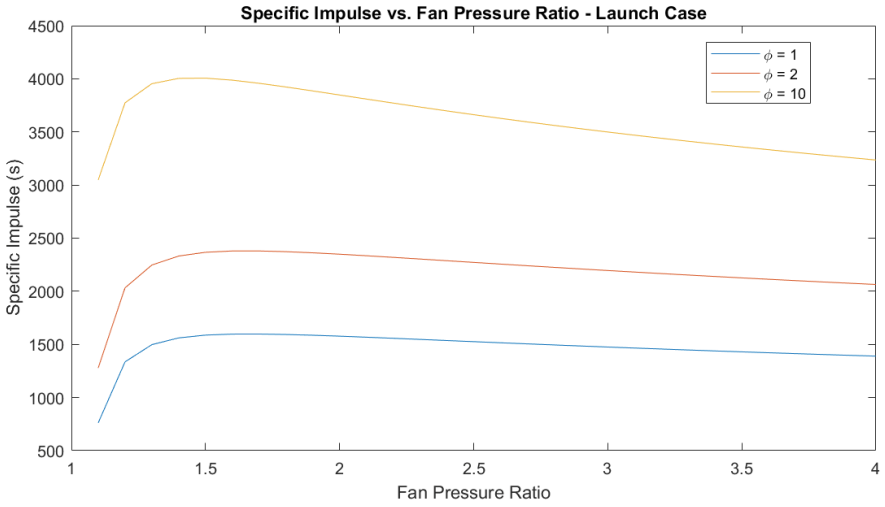
5.2 Analysis Tasks

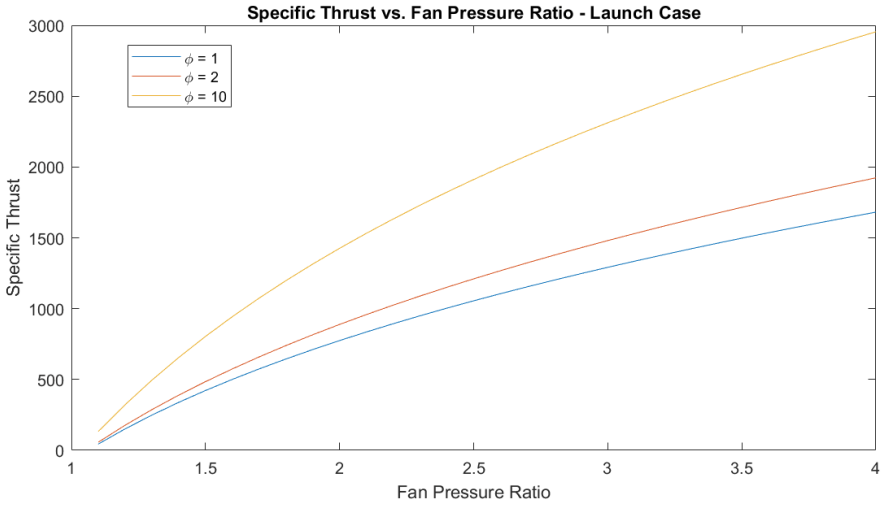
- Iterate on fan pressure ratio (FPR = 1.1 to 4.0)
- Iterate on capture area A0 to match thrust
- Use equivalence ratios φ = 1, 2, 10

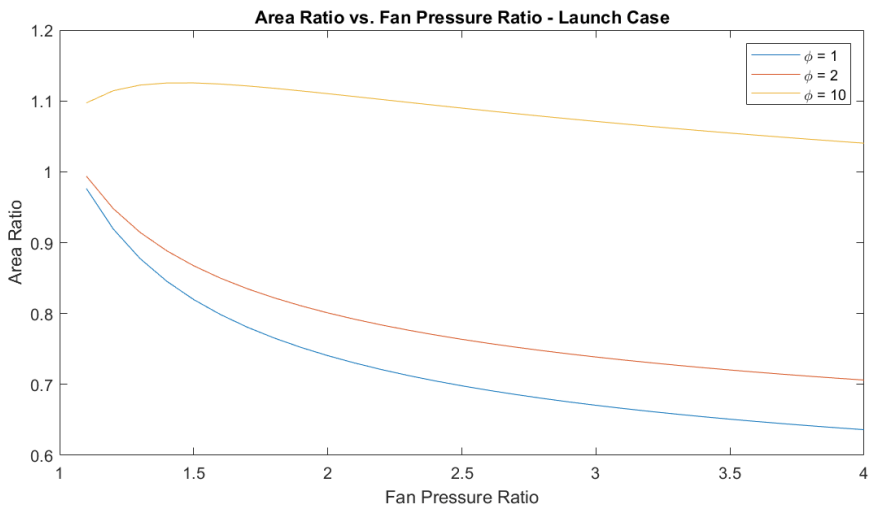
5.3 Required Plots and Data

Launch Case


SFC vs. Fan Pressure Ratio - Launch Case
SFC
Fan Pressure Ratio
ϕ = 1
ϕ = 2
ϕ = 10
5
4.5
4
3.5
3
2.5
2
1.5
1
0.5
1
1.5
2
2.5
3
3.5
4


Specific Impulse vs. Fan Pressure Ratio - Launch Case
Specific Impulse (s)
Fan Pressure Ratio
ϕ = 1
ϕ = 2
ϕ = 10
4500
4000
3500
3000
2500
2000
1500
1000
500
1
1.5
2
2.5
3
3.5
4


Specific Thrust vs. Fan Pressure Ratio - Launch Case
Specific Thrust
Fan Pressure Ratio
ϕ = 1
ϕ = 2
ϕ = 10
3000
2500
2000
1500
1000
500
0
1
1.5
2
2.5
3
3.5
4



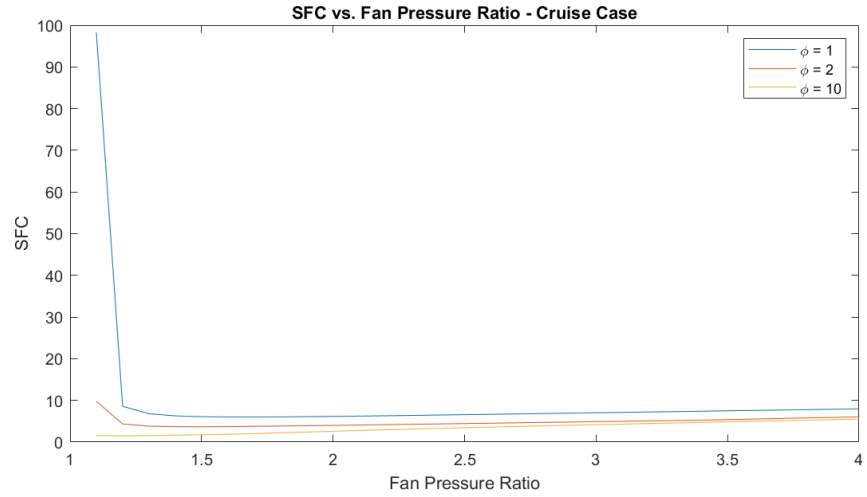
SFC vs. Fan pressure ratio (FPR)- The SFC decreases sharply as the FPR increase before flattening out to a slight increase at larger ratios all before 1.5. larger ϕ consistently show lower SFC which predicts that the ere is better propellant utilization. It can be predicted the moderate FPR values induce the best duel efficiency.

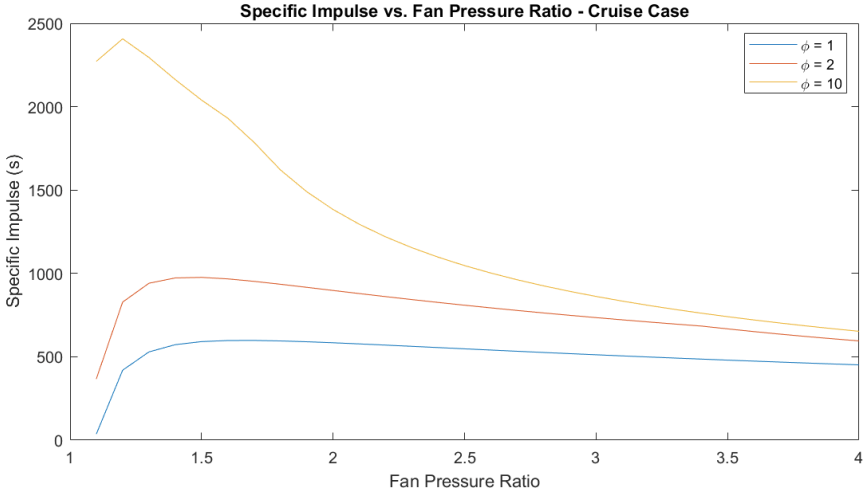
Specific Impulse vs. Fan pressure ratio (FPR)- The I_{sp} increase at low FPR and then flattens. The higher ϕ significantly increases the I_{sp} which reaches over 2000 seconds. This show that adding the fan enhances the energy efficiency while still diminishing flow constraint effects.

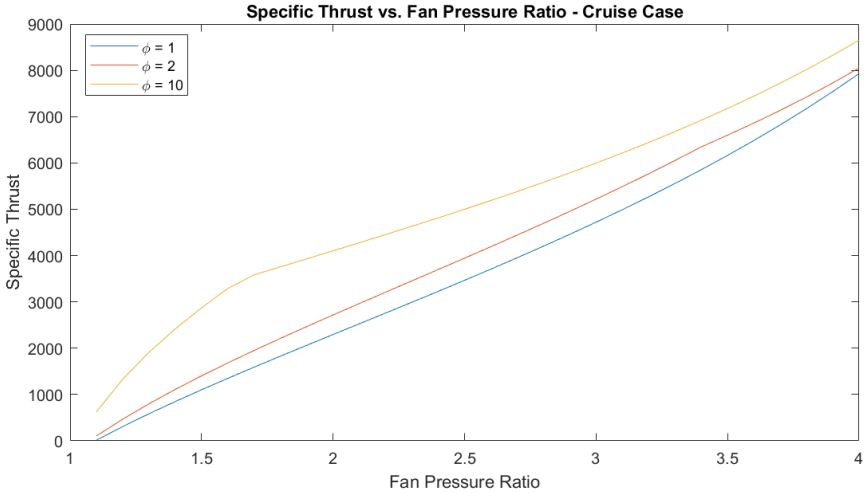
Specific thrust vs. Fan pressure ratio (FPR)- The thrust increases close to linearly as the FPR increases for all ϕ values. This suggest that the addition of the fan proportionally increases the thrust.

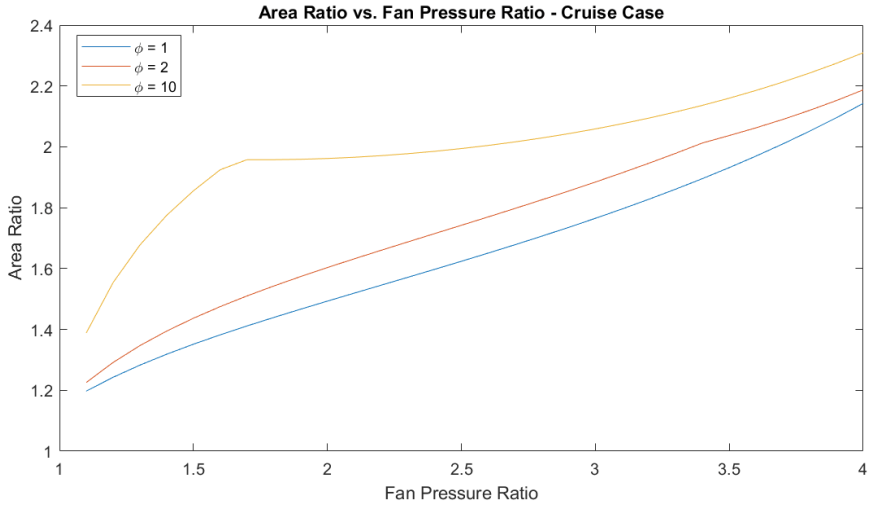
Area ratio vs. Fan Pressure ratio – The nozzle area ratio decrease with increasing FPR for ϕ= 1,2 while ϕ = 10 it remains close to or slightly above 1. This demonstrates that higher ϕ lessens the need for a large nozzle expansion as thrust is distributed more evenly.

Cruise Case


SFC vs. Fan Pressure Ratio - Cruise Case
SFC
Fan Pressure Ratio
ϕ = 1
ϕ = 2
ϕ = 10
0
10
20
30
40
50
60
70
80
90
100
1
1.5
2
2.5
3
3.5
4


Specific Impulse vs. Fan Pressure Ratio - Cruise Case
Specific Impulse (s)
Fan Pressure Ratio
ϕ = 1
ϕ = 2
ϕ = 10
0
500
1000
1500
2000
2500
1
1.5
2
2.5
3
3.5
4


Specific Thrust vs. Fan Pressure Ratio - Cruise Case
Specific Thrust
Fan Pressure Ratio
ϕ = 1
ϕ = 2
ϕ = 10
0
1000
2000
3000
4000
5000
6000
7000
8000
9000
1
1.5
2
2.5
3
3.5
4



SFC vs. Fan pressure ratio (FPR)- The SFC drops dramatically with increasing FPR before cutting to level across all ϕ with the highest of 10 having the lowest SFC. This suggest that even low FPR can significantly reduce fuel consumption at high speeds

Specific Impulse vs. Fan pressure ratio (FPR)- Specific impulse peaks at low FPR and then slowly decreases for all ϕ values. The highest I_{sp} occurring at low FPRs balances the effective combustion and expansion. However increasing the FPR beyond this point would cause slight efficiency losses.

Specific thrust vs. Fan pressure ratio (FPR)- Thrust increase steadily with FPR over al ϕ, with higher ratios generating more thrust. The growth is nearly linear suggesting that increasing the fan power consistently increase thrust at cruise.

Area ratio vs. Fan Pressure ratio – Required nozzle area ratio increases with the ϕ values, especially for the higher system. This reflects the need for a larger nozzle exit area to handle the increased total flow due to the fan. At cruise, ϕ=10 had an interesting diversion from normal the projects careful sizing to maximize efficiency and avoid under expansion .

5.4 Discussion Prompts

- *What is the maximum Isp and how does it compare to a pure rocket?*

The maximum Isp is approximately 4000 s for the launch case, and 2450 s for the cruise case. This means that the launch case is about 10x as efficient as a traditional hydrolox rocket, with its 400 s or so Isp, and that the cruise case is still about 6x more efficient than the pure rocket.

- *What happens at the Mach 0.8 condition?*

The Isp and specific thrust values increase massively in the Mach 0.8 condition compared to the Mach 4.0 condition, allowing for much more efficient flight. This suggests that the turbofan setup is much more effective in that situation compared to the high-speed, high-altitude Mach 4.0 condition.
- *How do area ratios and component geometries evolve with FPR? What if the geometry is not "Rubber"?*

The area ratio between the inlet and the nozzle increases with an increasing fan pressure ratio. This means that as the fan pressure ratio increases, the nozzle becomes increasingly larger in relation to the inlet. If the geometry was not rubber, this would mean that a specific fan pressure ratio would have to be chosen, most likely to optimize efficiency in a specific case, most likely cruising flight, and then, that geometry, would remain that same throughout the entire flight, likely resulting in efficiency losses in non-cruise flight conditions, where different geometry configurations would be more effective.

Table 2: Station properties for launch and cruise (FPR = 2, φ = 2).

See Appendix, or attached Google Sheet

https://docs.google.com/spreadsheets/d/13G1dATEa26tEEdhAg7wgPHcRKmOPR5iPEEfFGixbHek/edit?usp=sharing

6. Conclusion

- Summarize major trends in Isp, SFC, and specific thrust.

The specific impulse is starkly different for the two systems where the AAR has an increase in I_{sp} while the ATRR has a decrease before its plateau. The SFC decreases at varying slopes for all systems, and finally the specific thrust decreases exponentially in AAR system while it increases nearly linearly in the ATRR.
The main benefit of variable geometry is in its adaptability to different conditions. As seen from the data, in different conditions, different geometries and configurations are needed to maximize performance. This means that in an ideal world, fully variable engines would be the optimal way to design engines. However, in reality, designing engines with variable geometry is challenging, requiring complex mechanical systems that both cost much more money, and also increases the risk of failure. Thus, it's often better to simply find a "good-enough" configuration that provides acceptable performance, either by focusing on a deisgn that is most effective for the main flight condition that will be experienced, or by finding a configuration that provides alright, if not optimal performance across the entire range of conditions.

It is important to simplify assumptions and methods in the propulsion analysis to prevent calculations from becoming too complicated and tedious for what may or may not be a large increase in the accuracy of the calculations. Also, if certain conditions were not set, they

would have to be calculated or iterated over, adding more and more complication to the process.

Appendices

- MATLAB Code
- Tabulated Properties by Station
- Additional Graphs or Flow Visualizations

Appendices - Air Rocket Ramjet MATLAB Code

Main Function

```
function outputs = AirAugmentedRocketIterations(alt,A_0,phi)
%outputs = [SFCeng ISP FNWA A_no/A_0]
%Aerospace Propulsion - Semester Project
%Part 1 - Cruise Case (Mach 4.0, Altitude of 60000 ft)
%Compute m0 = m15 from A0.

%Variables (values being iterated)

%Enviromental Conditions (change between interations)
if alt == 60000
%Cruise Case
    M_0 = 4.0;  %[mach]
    ThrustTarget = 5000; %[lbf]
    Cd_cowl = 0.4;
    inletRecovery = 0.669;
    M_15 = 0.2;  %[mach]
else
%Launch Case
    M_0 = 0.8;  %[mach]
    ThrustTarget = 4000; %[lbf]
    Cd_cowl = 0.2;
    inletRecovery = 0.96;
    M_15 = 0.1;  %[mach]
end

%Constants - (remain same throughout all iterations)
gamma = 1.4;
Mwt_O2 = 32.00;
Mwt_N2 = 28.16;
Mwt_H2 = 2.00;
R_universal = 49710; %[ft*lbf/kmol*(slug/kg)*R]
delta_hf = 1.30E9;
a = 8.80E-5;
```

```
b = 540;
g0 = 32.17;
Cfg = 0.96;
massStep = 0.0001;

%Calculations from Freestream to Inlet
atmProperties = usAtm(alt);
P_0 = atmProperties(2);                    %[lbf/ft^2]
T_0 = atmProperties(1) + 459.67;                      %[R]
rho_0 = atmProperties(3);              %[slug/ft^3]
V_0 = M_0*atmProperties(4);                          %[ft/s]

Tt_0 = T_0*TrixM(M_0)^-1;
Pt_0 = P_0*PrixM(M_0)^-1;
q_0 = 0.5*rho_0*V_0^2;             %[lbf]
A_cowl = 0.1*A_0;                  %[ft]
mDot_0 = rho_0*V_0*A_0;
D_cowl = q_0*Cd_cowl*A_cowl;
D_ram = mDot_0*V_0;



phi_1 = 0;
Beta_1 = 3.762;

Pt_1 = Pt_0*inletRecovery;
Tt_1 = Tt_0;
fracs_1 = MassFracs(phi_1,Beta_1);
Mbp_1 = MWT_yHyOyN(fracs_1(1),fracs_1(2),fracs_1(3)); % Molecular Weight Total
R_1 = R_universal/Mbp_1;
Cp_1 = R_1*(gamma/(gamma-1));
md_1_H2 = fracs_1(1)*mDot_0;
md_1_O2 = fracs_1(2)*mDot_0;
md_1_N2 = fracs_1(3)*mDot_0;
md_1 = md_1_H2 + md_1_O2 + md_1_N2;

% pressures
Pt_15 = Pt_1;
P_15 = Pt_15*PrixM(M_15,gamma);

% temperatures
Tt_15 = Tt_1;
T_15 = Tt_15*TrixM(M_15,gamma);

% mass flows + fracs
phi_15 = phi_1;
Beta_15 = Beta_1;
```

```
Mbp_15 = Mbp_1;

md_15_H2 = md_1_H2;
md_15_O2 = md_1_O2;
md_15_N2 = md_1_N2;

md15 = md_15_H2 + md_15_O2 + md_15_N2;

% gas constants

R_15 = R_1;
Cp_15 = Cp_1;

% velocity
V_15 = M_15*sqrt(gamma*R_15*Tt_15);

% area
Rho_15 = P_15/(R_15*T_15);
A_15 = md15/(V_15*Rho_15);

% impulse
I_15 = md15*V_15 + P_15*A_15;

% enthalpy
ht_15 = Cp_15*Tt_15;

m_fuel = massStep;

while true

%Rocket Motor
Tt_ch = 540;
Pt_ch = 200000;
Cp_H2_ch = 87010;
Cp_O2_ch = 5438;
beta_ch = 0;
yONH_ch = MassFracs(phi,beta_ch);
a = 8.80*10^-5;
b = 540;
yH_ch = yONH_ch(1);
yO_ch = yONH_ch(2);
yN_ch = yONH_ch(3);

mDot_ch_H2 = yH_ch*m_fuel;
mDot_ch_O2 = yO_ch*m_fuel;
mDot_ch_N2 = yN_ch*m_fuel;
```

```
MWT_ch = MWT_yHyOyN(yH_ch,yO_ch,yN_ch);

R_ch = R_universal/MWT_ch;
Cp_ch = R_ch*(gamma/(gamma-1));
ht_ch = (1/m_fuel)*(mDot_ch_H2*Cp_H2_ch*Tt_ch + mDot_ch_O2*Cp_O2_ch*Tt_ch);

Tt_b = Ttbrn_yHyOyNhi(yH_ch, yO_ch, yN_ch, ht_ch, gamma);
n_ch = 1-a*(Tt_ch-b);

Tt_th = Tt_b;
Pt_th = Pt_ch;
M_th = 1;
mDot_th = m_fuel;
mft_th = MftxM(M_th);
A_th = (mDot_th*sqrt(R_ch*Tt_th))/(Pt_th*mft_th);

P_x = P_15;

M_x = MxPri((P_x/Pt_th));
T_x = TrixM(M_x)*Tt_th;
V_x = M_x*sqrt(gamma*R_ch*T_x);
A_x = ArixM(M_x)*A_th;
I_x = mDot_th*V_x + (P_x-P_0)*A_x;
Htx = (Cp_ch*Tt_th+V_x^2/2)*mDot_th;

%Station 7
% Flow rates [sl/s]
mdot_H2_7 = mDot_ch_H2 + md_1_H2;
mdot_O2_7 = mDot_ch_O2 + md_1_O2;
mdot_N2_7 = mDot_ch_N2 + md_1_N2;

% Feed total temperatures [R]
Tt_H2_7 = Tt_ch;
Tt_O2_7 = Tt_ch;
Tt_air_7 = Tt_1;

% Specific heats [ft^2/s^2/R]
Cp_H2_7 = Cp_H2_ch;
Cp_O2_7 = Cp_O2_ch;
Cp_air_7 = Cp_15;

%Total mass flow and mass fractions
mdot_7 = mdot_H2_7 + mdot_O2_7 + mdot_N2_7;
y_H_7 = mdot_H2_7 / mdot_7;
y_O_7 = mdot_O2_7 / mdot_7;
y_N_7 = mdot_N2_7 / mdot_7;
```

```
%Mixture MW and gas constant R
Mb_7 = MWT_yHyOyN(y_H_7,y_O_7,y_N_7);
R_7 = R_universal / Mb_7;                          % ft^2/s^2/R
Cp_7 = R_7*(gamma/(gamma-1));       % ft^2/s^2/R

%Initial enthalpy h_i
ht_7 = (1/mdot_7)*(mDot_ch_H2*Cp_H2_7*Tt_H2_7 +
mDot_ch_O2*Cp_O2_7*Tt_O2_7+md_1*Cp_air_7*Tt_air_7);

%Total temperature Tt7 after combustion
Tt_7 = Ttbrn_yHyOyNhi(y_H_7,y_O_7,y_N_7, ht_7, gamma);

eta_b_7 = 1-((Cp_7*Tt_7 - ht_7) / (mdot_H2_7 * delta_hf));

%Static flow quantities
A_7 = A_15 + A_x;
I_7 = I_x + I_15;
mfi_7 = Cp_7*Tt_7*(mdot_7/I_7)^2;
M_7 = MbxMfi(mfi_7);

T_7 = Tt_7*TrixM(M_7,gamma);
V_7 = M_7*sqrt(gamma*R_7*T_7);
rho_7 = mdot_7/(V_7*A_7);            % slug/ft^3
P_7 = rho_7*R_7*T_7;
Pt_7 = P_7 * (1 + (gamma - 1)/2 * M_7^2)^(gamma/(gamma - 1));  % psf

%Enthalpy and combustion efficiency
Ht_7 = (Cp_7*Tt_7+V_7^2/2)*mdot_7;


%Nozzle
R_no = R_7;
Cp_no = Cp_7;
md_no = mdot_7;
Tt_no = Tt_7;
Pt_no = Pt_7;
M_no = MxPri(P_0/Pt_no);
mft_no = MftxM(M_no);
A_no = (md_no*sqrt(R_no*Tt_no))/(Pt_no*mft_no);
T_no = Tt_no*TrixM(M_no,gamma);
P_no = Pt_no*PrixM(M_no,gamma);
V_no = M_no*sqrt(gamma*R_no*T_no);
F_no = md_no*V_no*Cfg;


%Gross Thrust
```

```
%F_gross = 18687 % from station 9
%% Ram Drag
%D_cowl = 268.3157;
%% Cowl (Additive) Drag
%D_ram = 1.3416e+04;

% Installed Thrust
F_installed = F_no - D_ram - D_cowl
if round(F_installed) >= ThrustTarget
    break
end
    m_fuel = m_fuel + massStep;
end

Cfg_no = (mdot_7 * V_7)/F_installed;
SFCeng = m_fuel*(32.174*3600)/F_installed
FNWA = F_installed/md_1 %Specific Thrust
ISP = 3600/SFCeng

outputs = [SFCeng ISP FNWA A_no/A_0];

%Verification Tables
format longG
Freestream = [alt; M_0; P_0; Pt_0; T_0; Tt_0; q_0; Cd_cowl; A_0; V_0; mDot_0;
A_cowl; D_cowl; D_ram];
Inlet = [inletRecovery; Pt_1; Tt_1; md_1; phi_1; Beta_1; md_1_H2; md_1_O2;
md_1_N2; Mbp_1; R_1; Cp_1];
Station15 =
[M_15;Pt_15;P_15;Tt_15;T_15;md15;md_15_H2;md_15_O2;md_15_N2;Mbp_15;R_15;Cp_15;
A_15;V_15;I_15;ht_15];
Chamber = [Tt_ch; Pt_ch; m_fuel; phi; beta_ch; Cp_H2_ch; Cp_O2_ch; mDot_ch_H2;
mDot_ch_O2; mDot_ch_N2; MWT_ch; R_ch; Cp_ch; ht_ch; n_ch; Tt_th];
Throat = A_th;
PrimaryExit = [M_x; P_x; T_x; V_x; A_x/A_th; A_x; I_x; Htx];
Station7 = [mdot_7; phi; mdot_H2_7; mdot_O2_7; mdot_N2_7; 0; Mb_7; R_7;
Cp_7;ht_7; eta_b_7; Tt_7; Ht_7; I_7; A_7; rho_7; V_7; T_7; M_7; P_7; Pt_7];
Nozzle = [Mb_7; R_no; Cp_no; Tt_no; T_no;
Pt_no;P_no;M_no;A_no;V_no;md_no;Cfg;F_no];
Performance = [F_installed; SFCeng;FNWA;ISP];

function resultsTable = usAtm(alt)
%Outputs T in F, P in psf, densitry in slug/ft^3, and speed of sound it
%ft/s)
atmoData = [ -5000  76.84  17.55427.45
0      59     14.696 23.77
5000   41.17  12.228 20.48
10000  23.36  10.108 17.56
```

```
15000  5.55   8.297  14.96
20000  -12.26 6.759  12.67
25000  -30.05 5.461  10.66
30000  -47.83 4.373  8.91
35000  -65.61 3.468  7.38
40000  -69.7  2.73   5.87
45000  -69.7  2.149  4.62
50000  -69.7  1.692  3.64
60000  -69.7  1.049  2.26
70000  -67.42 0.651  1.39
80000  -61.98 0.406  0.86
90000  -56.54 0.255  0.56
100000 -51.1  0.162  0.33
150000 19.4   0.02   0.037
200000 -19.78 0.003  0.0053
250000 -88.77 0      0.00065];


t = interp1(atmoData(:,1),atmoData(:,2),alt);

p = interp1(atmoData(:,1),atmoData(:,3),alt);

p = p*144;

d = interp1(atmoData(:,1),atmoData(:,4),alt)*10^-4;

s = sqrt(1.4*1715*(t + 459.6));

resultsTable = [t p d s];

end


function [ yOHN ] = MassFracs( Phi, Beta )
% Computes the mass fractions of oxygen, hydrogen, and nitrogen atoms
% in the mixture.
% Inputs: Phi  = mixture equivalence ratio
%         Beta = molar ratio of nitrogen to oxygen atoms in the mixture
% Output: Yh = mass fraction of hydrogen atoms
%
% Assumed molecular weights:
%         O  = 16
%         H  =  1
%         N  = 14.082
%             (adjusted to account for nitrogen, argon, etc. in air.)
```

```
% Set the number of moles of each component, assuming one mole of oxygen
atoms.
  nO = 1;
  nN = Beta * nO;
  nH = 2 * Phi * nO;

% Calculate the resulting mixture mass and the mass fration of hydrogen atoms
  mMix = 16 * nO + 14.082 * nN + 1 * nH;
  yO   = (16 * nO) ./ mMix;
  yH   = (1 * nH) ./ mMix;
  yN   = (14.082 * nN) ./ mMix;

  yOHN = [ yH, yO, yN ];
end

function [ MWT ] = MWT_yHyOyN(yH, yO, yN)
% Computes the mixture average molecular weight assuming the hydrogen-oxygen
reaction goes to completion
% Inputs: yH = mass fraction of hydrogen atoms
%         yO = mass fraction of oxygen atoms
%         yN = mass fraction of inert atoms
%
% Assumed molecular weights:
%         O2  = 32
%         H2  =  2
%         H2O = 18
%         N2  = 28.164
%              (adjusted to account for nitrogen, argon, etc. in air.)

% Compute the equivalence ratio and then, for one half mole of O2 reactant,
% find the number of moles of each resulting consitutuent including water,
% and unburned hydrogen, oxygen, and nitrogen (inerts).
  Phi = 8 * (yH ./ yO);

  nH2O = min(Phi, 1);
  nH2u = max(Phi - 1, 0);
  nO2u = 0.5 * max(1 - Phi, 0);
  nN2  = 0.5 * (yN / 28.164) ./ (yO / 32);

% Find the mixture weight and average molecular weight.
  mH2O = 18 * nH2O;
  mH2u = 2 * nH2u;
  mO2u = 32 * nO2u;
  mN2  = 28.164 * nN2;

  MWT = (mH2O + mH2u + mO2u + mN2) ./ (nH2O + nH2u + nO2u + nN2);
```

```
end

function [Tri] = TrixM(M, g)
% Computes T/Tt for the given M.
  if nargin < 2, g = 1.4; end
  Tri = 1 / (1 + (g - 1) / 2 * M ^ 2);
end

function [Pri] = PrixM(M, g)
% Computes P/Pt for the given M.
  if nargin < 2, g = 1.4; end
  Pri = (1 + (g - 1) / 2 * M ^ 2) ^ (g / (1 - g));
end

function [ Ttbrn ] = Ttbrn_yHyOyNhi(yH, yO, yN, hi, g)
% Approximates the gas total temperature after combution of hydrogen
% including a crude estimate to account for dissociation.
% Inputs: yH  = mass fraction of hydrogen atoms
%         yO  = mass fraction of oxygen atoms
%         yN  = mass fraction of inert atoms
%         hi  = specific enthalpy of reactants, not including heats of
formation [ft^2/s^2]
%
% 'Combustion of hydrogen approximated by:
%         hpr = heat of combustion = 1.3e9 [ft^2/s^2]
%         a   = combustion efficiency/dissociation sensitivity = 8.8e-5
%         b   = combustion efficiency/dissociation constant = 540

% Set constants
  hpr = 1300000000;
  a = 0.000088;
  b = 540;

% Ensure mass fractions add to one and get the combusted gas properties.
  Ytot = yH + yO + yN;
  yH = yH / Ytot;
  yO = yO / Ytot;
  yN = yN / Ytot;

  MWT = MWT_yHyOyN(yH, yO, yN);
  R = 49710. / MWT;
  Cp = R * g / (g - 1);

% Compute the mass fraction of burned hydrogen and the mixture temperature.
  Yh2b = min(yH, yO / 8);

  Ttbrn = (hi + Yh2b * hpr * (1 + a * b)) / (Cp + a * Yh2b * hpr);
```

```
end

function [Mft] = MftxM(M, g)
% Computes mdot(RTt)^.5/(PtA) for the given M.
  if nargin < 2, g = 1.4; end
  Mft = g ^ 0.5 * M * (1 + (g - 1) / 2 * M ^ 2) ^ (-(g + 1) / (2 * (g - 1)));
end

function [M] = MxPri(PR, g)
% Computes M for the given P/Pt.
  if nargin < 2, g = 1.4; end
  M = (2 / (g - 1) * (PR ^ ((1 - g) / g) - 1)) ^ 0.5;
end

function [Ari] = ArixM(M, g)
% Computes A/A* for the given M.
  if nargin < 2, g = 1.4; end
  Ari = (2 * (1 + (g - 1) / 2 * M ^ 2) / (g + 1)) ^ ((g + 1) / (2 * (g - 1)))
/ M;
end

function [Mfi] = MfixM(M, g)
% Computes ht*(mdot/I)^2 for the given M.
  if nargin < 2, g = 1.4; end
  Mfi = g ^ 2 / (g - 1) * M ^ 2 * (1 + (g - 1) / 2 * M ^ 2) / (1 + g * M ^ 2)
^ 2;
end

function [Mb] = MbxMfi(MFIin, g)
% Computes the subsonic Mach number for the given mass flow function,
ht*(mdot/I)^2

  if nargin < 2, g = 1.4; end
  eps = 0.0000001;
  itmax = 1000;

  Ma = 0;
  Fa = MfixM(Ma, g);
  Mb = 1;
  Fb = MfixM(Mb, g);

  for iter = 1: itmax
    if mod(iter, 2) == 1
      Mg = 0.5 * (Ma + Mb);
    else
      Mg = Ma + (sqrt(MFIin) - sqrt(Fa)) * (Mb - Ma) / (sqrt(Fb) - sqrt(Fa));
```

```
    end
    Fg = MfixM(Mg, g);

    if abs(Fg - MFIin) <= eps
      break;
    elseif Fg > MFIin
      Mb = Mg;
      Fb = Fg;
    else
      Ma = Mg;
      Fa = Fg;
    end
  end

  Mb = Mg;
end




end
```

Launch Case Iteration

```
clc
close all
clear all


A_0s = 0:0.5:15;
A_0s(1) = 1*10^-4;
phis = [1 2 10 1000];

SFCs = zeros(length(A_0s),length(phis));
ISPs = zeros(length(A_0s),length(phis));
SpecificThrusts = zeros(length(A_0s),length(phis));
AreaRatios = zeros(length(A_0s),length(phis));

for i = 1:length(A_0s)
    for j = 1:length(phis)
        output = AirAugmentedRocketIterations(30000,A_0s(i),phis(j));
        SFCs(i,j) = output(1);
        ISPs(i,j) = output(2);
        SpecificThrusts(i,j) = output(3);
        AreaRatios(i,j) = output(4);
    end
```

```
end

plot(A_0s,SFCs)
title("SFC vs. Inlet Area - Launch Case")
xlabel("Inlet Area (ft^2)")
ylabel("SFC")
legend('\phi = 1','\phi = 2', '\phi = 100', '\phi = 1000', 'Location',
'NorthEast')

figure
plot(A_0s,ISPs)
title("Specific Impulse vs. Inlet Area - Launch Case")
xlabel("Inlet Area (ft^2)")
ylabel("Specific Impulse (s)")
legend('\phi = 1','\phi = 2', '\phi = 100', '\phi = 1000', 'Location',
'NorthWest')

figure
plot(A_0s(2:length(A_0s)),SpecificThrusts(2:length(A_0s),:))
title("Specific Thrust vs. Inlet Area - Launch Case")
xlabel("Inlet Area (ft^2)")
ylabel("Specific Thrust")
legend('\phi = 1','\phi = 2', '\phi = 100', '\phi = 1000', 'Location',
'NorthEast')

figure
plot(A_0s(2:length(A_0s)),AreaRatios(2:length(A_0s),:))
title("Area Ratio vs. Inlet Area - Launch Case")
xlabel("Inlet Area (ft^2)")
ylabel("Area Ratio")
legend('\phi = 1','\phi = 2', '\phi = 100', '\phi = 1000', 'Location',
'NorthEast')
```

Cruise Case Iteration

```
clc
close all
clear all

A_0s = 0:0.5:15;
A_0s(1) = 1*10^-4;
phis = [1 2 10 1000]

SFCs = zeros(length(A_0s),length(phis));
ISPs = zeros(length(A_0s),length(phis));
```

```
SpecificThrusts = zeros(length(A_0s),length(phis));
AreaRatios = zeros(length(A_0s),length(phis));

for i = 1:length(A_0s)
    for j = 1:length(phis)
        output = AirAugmentedRocketIterations(60000,A_0s(i),phis(j));
        SFCs(i,j) = output(1);
        ISPs(i,j) = output(2);
        SpecificThrusts(i,j) = output(3);
        AreaRatios(i,j) = output(4);
    end
end

plot(A_0s,SFCs)
title("SFC vs. Inlet Area - Cruise Case")
xlabel("Inlet Area (ft^2)")
ylabel("SFC")
legend('\phi = 1','\phi = 2', '\phi = 100', '\phi = 1000', 'Location',
'NorthEast')

figure
plot(A_0s,ISPs)
title("Specific Impulse vs. Inlet Area - Cruise Case")
xlabel("Inlet Area (ft^2)")
ylabel("Specific Impulse (s)")
legend('\phi = 1','\phi = 2', '\phi = 100', '\phi = 1000', 'Location',
'NorthWest')

figure
plot(A_0s(2:length(A_0s)),SpecificThrusts(2:length(A_0s),:))
title("Specific Thrust vs. Inlet Area - Cruise Case")
xlabel("Inlet Area (ft^2)")
ylabel("Specific Thrust")
legend('\phi = 1','\phi = 2', '\phi = 100', '\phi = 1000', 'Location',
'NorthEast')

figure
plot(A_0s(2:length(A_0s)),AreaRatios(2:length(A_0s),:))
title("Area Ratio vs. Inlet Area - Cruise Case")
xlabel("Inlet Area (ft^2)")
ylabel("Area Ratio")
legend('\phi = 1','\phi = 2', '\phi = 100', '\phi = 1000', 'Location',
'NorthEast')
```

Appendices - Air Turbo Rocket Ramjet

Main Function

```
function outputs = AirTurboRocketRamjetIterations(alt,PI_f,phi)
%Aerospace Propulsion - Semester Project
%Part 1 - Cruise Case (Mach 4.0, Altitude of 60000 ft)
%Compute m0 = m15 from A0.

%Enviromental Conditions (change between interations)
if alt == 60000
%Cruise Case
    M_0 = 4.0;  %[mach]
    ThrustTarget = 5000; %[lbf]
    Cd_cowl = 0.4
    inletRecovery = 0.669;
    M_15 = 0.2;  %[mach]
else
%Launch Case
    M_0 = 0.8;  %[mach]
    ThrustTarget = 4000; %[lbf]
    Cd_cowl = 0.2
    inletRecovery = 0.96;
    M_15 = 0.2;  %[mach]
end

%Constants - (remain same throughout all iterations)
gamma = 1.4;
Mwt_O2 = 32.00;
Mwt_N2 = 28.16;
Mwt_H2 = 2.00;
R_universal = 49710; %[ft*lbf/kmol*(slug/kg)*R]
delta_hf = 1.30E9;
a = 8.80E-5;
b = 540;
g0 = 32.17;
Cfg = 0.96;


areaStep = 0.01;
A_0 = areaStep;


%Calculations from Freestream to Inlet
atmProperties = usAtm(alt);
P_0 = atmProperties(2);                    %[lbf/ft^2]
T_0 = atmProperties(1) + 459.67;                    %[R]
```

```
rho_0 = atmProperties(3);              %[slug/ft^3]
V_0 = M_0*atmProperties(4);                        %[ft/s]

Tt_0 = T_0*(1+((gamma-1)*M_0^2)/2);
Pt_0 = P_0*(Tt_0/T_0)^(gamma/(gamma-1));
q_0 = 0.5*rho_0*V_0^2;             %[lbf]

while true

A_cowl = 0.1*A_0;                  %[ft]
mDot_0 = rho_0*V_0*A_0;
D_cowl = q_0*Cd_cowl*A_cowl;
D_ram = mDot_0*V_0;



phi_1 = 0;
Beta_1 = 3.762;

Pt_1 = Pt_0*inletRecovery;
Tt_1 = Tt_0;
fracs_1 = MassFracs(phi_1,Beta_1);
Mbp_1 = MWT_yHyOyN(fracs_1(1),fracs_1(2),fracs_1(3)); % Molecular Weight Total
R_1 = R_universal/Mbp_1
Cp_1 = R_1*(gamma/(gamma-1))
md_1_H2 = fracs_1(1)*mDot_0;
md_1_O2 = fracs_1(2)*mDot_0;
md_1_N2 = fracs_1(3)*mDot_0;
md_1 = md_1_H2 + md_1_O2 + md_1_N2;

% Fan
eta_f = 0.95;
R_fan = R_1;
Cp_fan = Cp_1;

Pt_fan = PI_f * Pt_1;
Tt_fan = Tt_1*(1+(1/eta_f)*(PI_f^((gamma-1)/gamma)-1));
W_fan = md_1*Cp_fan*(Tt_fan-Tt_1)

%Station 15
% pressures
Pt_15 = Pt_fan;
P_15 = Pt_15*PrixM(M_15,gamma);

% temperatures
Tt_15 = Tt_fan;
T_15 = Tt_15*TrixM(M_15,gamma);
```

```
% mass flows + fracs
phi_15 = phi_1;
Beta_15 = Beta_1;

Mbp_15 = Mbp_1;

md_15_H2 = md_1_H2;
md_15_O2 = md_1_O2;
md_15_N2 = md_1_N2;

md15 = md_15_H2 + md_15_O2 + md_15_N2;

% gas constants

R_15 = R_fan;
Cp_15 = Cp_fan;

% velocity
V_15 = M_15*sqrt(gamma*R_15*Tt_15);

% area
Rho_15 = P_15/(R_15*T_15);
A_15 = md15/(V_15*Rho_15);

% impulse
I_15 = md15*V_15 + P_15*A_15;

% enthalpy
ht_15 = md15*Cp_15*Tt_15;

%Rocket Motor
Tt_ch = 540;
Pt_ch = 200000;
Cp_H2_ch = 87010;
Cp_O2_ch = 5438;
beta_ch = 0;
m_fuel = 1;
yONH_ch = MassFracs(phi,beta_ch);
a = 8.80*10^-5;
b = 540;
yH_ch = yONH_ch(1);
yO_ch = yONH_ch(2);
yN_ch = yONH_ch(3);

mDot_ch_H2 = yH_ch*m_fuel;
mDot_ch_O2 = yO_ch*m_fuel;
mDot_ch_N2 = yN_ch*m_fuel;
```

```
MWT_ch = MWT_yHyOyN(yH_ch,yO_ch,yN_ch)

R_ch = R_universal/MWT_ch
Cp_ch = R_ch*(gamma/(gamma-1))
ht_ch = (1/m_fuel)*(mDot_ch_H2*Cp_H2_ch*Tt_ch + mDot_ch_O2*Cp_O2_ch*Tt_ch)

Tt_b = Ttbrn_yHyOyNhi(yH_ch, yO_ch, yN_ch, ht_ch, gamma)
n_b = 1-a*(Tt_b-b)

mch_brnH2 = 69420;  % DONT ACC NEED THIS

% Throat, dont care
Tt_th = Tt_b
Pt_th = Pt_ch
M_th = 1
mDot_th = m_fuel
mft_th = MftxM(M_th)
A_th = (mDot_th*sqrt(R_ch*Tt_th))/(Pt_th*mft_th)

% turbine exit wahoo!


M_x = 1;

P_x = P_15;
Pt_x = (1/PrixM(M_x))*P_x;

eta_tau = 0.9;
eta_mu = 0.99;



PI_T = Pt_x/Pt_ch;
tau = 1-eta_tau*(1-PI_T^((gamma-1)/gamma));
Tt_x = tau*Tt_b;
SpecificTurbinePower = Cp_ch*(Tt_b-Tt_x)
md_x = W_fan/(eta_mu*SpecificTurbinePower)
T_x = TrixM(M_x)*Tt_x;

beta_x = beta_ch;
phi_x = phi;
yONH_x = MassFracs(phi_x,beta_x);

yH_x = yONH_x(1);
yO_x = yONH_x(2);
yN_x = yONH_x(3);
```

```
mDot_x_H2 = yH_x*md_x;
mDot_x_O2 = yO_x*md_x;
mDot_x_N2 = yN_x*md_x;

MWT_x = MWT_yHyOyN(yH_x,yO_x,yN_x);

R_x = R_universal/MWT_x;
Cp_x = R_x*(gamma/(gamma-1));
rho_x = (P_x*MWT_x)/(R_universal*T_x);
U_x = M_x*sqrt(gamma*R_x*T_x);
A_x = md_x/(rho_x*U_x);
I_x = md_x*U_x + (P_x-P_0)*A_x;
dh = ht_ch+(Cp_x*Tt_x+U_x^2/2)*md_x;



%Station 7
% Flow rates [sl/s]
mdot_H2_7 = mDot_x_H2 + md_1_H2;
mdot_O2_7 = mDot_x_O2 + md_1_O2;
mdot_N2_7 = mDot_x_N2 + md_1_N2;

% Feed total temperatures [R]
Tt_H2_7 = Tt_ch;
Tt_O2_7 = Tt_ch;
Tt_air_7 = Tt_1;

% Specific heats [ft^2/s^2/R]
Cp_H2_7 = Cp_H2_ch;
Cp_O2_7 = Cp_O2_ch;
Cp_air_7 = Cp_15;

%Total mass flow and mass fractions
mdot_7 = mdot_H2_7 + mdot_O2_7 + mdot_N2_7;
y_H_7 = mdot_H2_7 / mdot_7;
y_O_7 = mdot_O2_7 / mdot_7;
y_N_7 = mdot_N2_7 / mdot_7;

%Mixture MW and gas constant R
Mb_7 = MWT_yHyOyN(y_H_7,y_O_7,y_N_7)
R_7 = R_universal / Mb_7;                          % ft^2/s^2/R
Cp_7 = R_7*(gamma/(gamma-1))       % ft^2/s^2/R

%Initial enthalpy h_i
ht_7 = (1/mdot_7)*(mDot_x_H2*Cp_H2_7*Tt_H2_7 +
mDot_x_O2*Cp_O2_7*Tt_O2_7+md_1*Cp_air_7*Tt_air_7)
```

```
%Total temperature Tt7 after combustion
Tt_7 = Ttbrn_yHyOyNhi(y_H_7,y_O_7,y_N_7, ht_7, gamma)

eta_b_7 = 1-((Cp_7*Tt_7 - ht_7) / (mdot_H2_7 * delta_hf));

%Static flow quantities
%rho_x = (P_x*MWT_x)/(R_universal*T_x);
% rho_7 = P_static_7 / (R_7 * T_static_7);            % slug/ft^3
%
%        % ft/s
% A_7 = A_15+A_x;                 % ft^2
% M_7 = V_7 / sqrt(gamma * R_7 * T_static_7);
% Pt_7 = P_static_7 * (1 + (gamma - 1)/2 * M_7^2)^(gamma/(gamma - 1));  % psf
% T_7 = Tt_7*TrixM(M_7,gamma);
% P_7 = Pt_7*PrixM(M_7,gamma);
% V_7 = sqrt(2 * mdot_H2_7 * delta_hf * eta_b_7)/10;

A_7 = A_15 + A_x
I_7 = I_x + I_15
mfi_7 = Cp_7*Tt_7*(mdot_7/I_7)^2
M_7 = MbxMfi(mfi_7)

T_7 = Tt_7*TrixM(M_7,gamma)
V_7 = M_7*sqrt(gamma*R_7*T_7)
rho_7 = mdot_7/(V_7*A_7);            % slug/ft^3
P_7 = rho_7*R_7*T_7
Pt_7 = P_7 * (1 + (gamma - 1)/2 * M_7^2)^(gamma/(gamma - 1));  % psf

%Enthalpy and combustion efficiency
Ht_7 = (Cp_7*Tt_7+V_7^2/2)*mdot_7
%I_7 = mdot_7*V_7 + (P_7-P_0)*A_7

m7_brnH2 = 69420;  % DONT ACC NEED THIS

%Nozzle
R_no = R_7
Cp_no = Cp_7
md_no = mdot_7
Tt_no = Tt_7
Pt_no = Pt_7
M_no = MxPri(P_0/Pt_no)
mft_no = MftxM(M_no)
A_no = (md_no*sqrt(R_no*Tt_no))/(Pt_no*mft_no)
T_no = Tt_no*TrixM(M_no,gamma);
P_no = Pt_no*PrixM(M_no,gamma);
```

```
V_no = M_no*sqrt(gamma*R_no*T_no)
F_no = md_no*V_no*Cfg



format longG
Freestream = [alt; M_0; P_0; Pt_0; T_0; Tt_0; q_0; Cd_cowl; A_0; V_0; mDot_0;
A_cowl; D_cowl; D_ram]
Inlet = [inletRecovery; Pt_1; Tt_1; md_1; phi_1; Beta_1; md_1_H2; md_1_O2;
md_1_N2; Mbp_1; R_1; Cp_1]
Station15 = [M_15;Pt_15;P_15;Tt_15;T_15;md15;A_15;V_15;I_15;
ht_15;md_15_H2;md_15_O2;md_15_N2;Mbp_15;R_15;Cp_15]
Chamber = [Tt_ch; Pt_ch; m_fuel; phi; beta_ch; Cp_H2_ch; Cp_O2_ch; mDot_ch_H2;
mDot_ch_O2; mDot_ch_N2; MWT_ch; R_ch; Cp_ch; ht_ch; n_b; Tt_th]
Throat = A_th
TurbineExit = [eta_tau; eta_mu; M_x; P_x; Pt_x; T_x; Tt_x; dh; md_x; phi_x;
beta_x; mDot_x_H2; mDot_x_O2; mDot_x_N2; MWT_x; R_x; Cp_x; U_x; A_x; I_x]
Station7 = [mdot_7; mdot_H2_7; mdot_O2_7; mdot_N2_7; Mb_7; R_7; Cp_7; eta_b_7;
Tt_7; Ht_7; I_7; A_7; rho_7; V_7; T_7; M_7; P_7; Pt_7]
Nozzle = [Mb_7; R_no; Cp_no; Tt_no; T_no;
Pt_no;P_no;M_no;A_no;V_no;md_no;Cfg;F_no]

Tt_0 = T_0*TrixM(M_0)^-1;

Pt_0 = P_0*PrixM(M_0)^-1;

q_0 = 0.5*rho_0*V_0^2;              %[lbf]

A_cowl = 0.1*A_0;                   %[ft]

mDot_0 = rho_0*V_0*A_0;

D_cowl = q_0*Cd_cowl*A_cowl;

D_ram = mDot_0*V_0;


% Installed Thrust

F_installed = F_no - D_ram - D_cowl;
if round(F_installed) >= ThrustTarget
    break
end
    A_0 = A_0 + areaStep;
end
```

```
Cfg_no = (mdot_7 * V_7)/F_installed;

SFCeng = md_x*(32.174*3600)/F_installed

FNWA = F_installed/md_1 %Specific Thrust

ISP = 3600/SFCeng

FN_AMX = ThrustTarget/A_0

outputs = [SFCeng ISP FNWA A_no/A_0];

function resultsTable = usAtm(alt)
%Outputs T in F, P in psf, densitry in slug/ft^3, and speed of sound it
%ft/s)
atmoData = [ -5000  76.84  17.55427.45
0      59     14.696 23.77
5000   41.17  12.228 20.48
10000  23.36  10.108 17.56
15000  5.55   8.297  14.96
20000  -12.26 6.759  12.67
25000  -30.05 5.461  10.66
30000  -47.83 4.373  8.91
35000  -65.61 3.468  7.38
40000  -69.7  2.73   5.87
45000  -69.7  2.149  4.62
50000  -69.7  1.692  3.64
60000  -69.7  1.049  2.26
70000  -67.42 0.651  1.39
80000  -61.98 0.406  0.86
90000  -56.54 0.255  0.56
100000 -51.1  0.162  0.33
150000 19.4   0.02   0.037
200000 -19.78 0.003  0.0053
250000 -88.77 0      0.00065];

t = interp1(atmoData(:,1),atmoData(:,2),alt);

p = interp1(atmoData(:,1),atmoData(:,3),alt);

p = p*144;

d = interp1(atmoData(:,1),atmoData(:,4),alt)*10^-4;
```

```
s = sqrt(1.4*1715*(t + 459.6));

resultsTable = [t p d s];
end

function [ yOHN ] = MassFracs( Phi, Beta )
% Computes the mass fractions of oxygen, hydrogen, and nitrogen atoms
% in the mixture.
% Inputs: Phi  = mixture equivalence ratio
%         Beta = molar ratio of nitrogen to oxygen atoms in the mixture
% Output: Yh = mass fraction of hydrogen atoms
%
% Assumed molecular weights:
%         O  = 16
%         H  =  1
%         N  = 14.082
%             (adjusted to account for nitrogen, argon, etc. in air.)

% Set the number of moles of each component, assuming one mole of oxygen
atoms.
  nO = 1;
  nN = Beta * nO;
  nH = 2 * Phi * nO;

% Calculate the resulting mixture mass and the mass fration of hydrogen atoms
  mMix = 16 * nO + 14.082 * nN + 1 * nH;
  yO   = (16 * nO) ./ mMix;
  yH   = (1 * nH) ./ mMix;
  yN   = (14.082 * nN) ./ mMix;

  yOHN = [ yH, yO, yN ];
end

function [ MWT ] = MWT_yHyOyN(yH, yO, yN)
% Computes the mixture average molecular weight assuming the hydrogen-oxygen
reaction goes to completion
% Inputs: yH = mass fraction of hydrogen atoms
%         yO = mass fraction of oxygen atoms
%         yN = mass fraction of inert atoms
%
% Assumed molecular weights:
%         O2  = 32
%         H2  =  2
%         H2O = 18
%         N2  = 28.164
%              (adjusted to account for nitrogen, argon, etc. in air.)
```

```
% Compute the equivalence ratio and then, for one half mole of O2 reactant,
% find the number of moles of each resulting consitutuent including water,
% and unburned hydrogen, oxygen, and nitrogen (inerts).
  Phi = 8 * (yH ./ yO);

  nH2O = min(Phi, 1);
  nH2u = max(Phi - 1, 0);
  nO2u = 0.5 * max(1 - Phi, 0);
  nN2  = 0.5 * (yN / 28.164) ./ (yO / 32);

% Find the mixture weight and average molecular weight.
  mH2O = 18 * nH2O;
  mH2u = 2 * nH2u;
  mO2u = 32 * nO2u;
  mN2  = 28.164 * nN2;

  MWT = (mH2O + mH2u + mO2u + mN2) ./ (nH2O + nH2u + nO2u + nN2);

end

function [Tri] = TrixM(M, g)
% Computes T/Tt for the given M.
  if nargin < 2, g = 1.4; end
  Tri = 1 / (1 + (g - 1) / 2 * M ^ 2);
end

function [Pri] = PrixM(M, g)
% Computes P/Pt for the given M.
  if nargin < 2, g = 1.4; end
  Pri = (1 + (g - 1) / 2 * M ^ 2) ^ (g / (1 - g));
end

function [ Ttbrn ] = Ttbrn_yHyOyNhi(yH, yO, yN, hi, g)
% Approximates the gas total temperature after combution of hydrogen
% including a crude estimate to account for dissociation.
% Inputs: yH  = mass fraction of hydrogen atoms
%         yO  = mass fraction of oxygen atoms
%         yN  = mass fraction of inert atoms
%         hi  = specific enthalpy of reactants, not including heats of
formation [ft^2/s^2]
%
% 'Combustion of hydrogen approximated by:
%         hpr = heat of combustion = 1.3e9 [ft^2/s^2]
%         a   = combustion efficiency/dissociation sensitivity = 8.8e-5
%         b   = combustion efficiency/dissociation constant = 540

% Set constants
```

```
  hpr = 1300000000;
  a = 0.000088;
  b = 540;

% Ensure mass fractions add to one and get the combusted gas properties.
  Ytot = yH + yO + yN;
  yH = yH / Ytot;
  yO = yO / Ytot;
  yN = yN / Ytot;

  MWT = MWT_yHyOyN(yH, yO, yN);
  R = 49710. / MWT;
  Cp = R * g / (g - 1);

% Compute the mass fraction of burned hydrogen and the mixture temperature.
  Yh2b = min(yH, yO / 8);

  Ttbrn = (hi + Yh2b * hpr * (1 + a * b)) / (Cp + a * Yh2b * hpr);

end

function [Mft] = MftxM(M, g)
% Computes mdot(RTt)^.5/(PtA) for the given M.
  if nargin < 2, g = 1.4; end
  Mft = g ^ 0.5 * M * (1 + (g - 1) / 2 * M ^ 2) ^ (-(g + 1) / (2 * (g - 1)));
end

function [M] = MxPri(PR, g)
% Computes M for the given P/Pt.
  if nargin < 2, g = 1.4; end
  M = (2 / (g - 1) * (PR ^ ((1 - g) / g) - 1)) ^ 0.5;
end

function [Ari] = ArixM(M, g)
% Computes A/A* for the given M.
  if nargin < 2, g = 1.4; end
  Ari = (2 * (1 + (g - 1) / 2 * M ^ 2) / (g + 1)) ^ ((g + 1) / (2 * (g - 1)))
/ M;
end

function [Mb] = MbxMfi(MFIin, g)
% Computes the subsonic Mach number for the given mass flow function,
ht*(mdot/I)^2

  if nargin < 2, g = 1.4; end
  eps = 0.0000001;
  itmax = 1000;
```

```
  Ma = 0;
  Fa = MfixM(Ma, g);
  Mb = 1;
  Fb = MfixM(Mb, g);

  for iter = 1: itmax
    if mod(iter, 2) == 1
      Mg = 0.5 * (Ma + Mb);
    else
      Mg = Ma + (sqrt(MFIin) - sqrt(Fa)) * (Mb - Ma) / (sqrt(Fb) - sqrt(Fa));
    end
    Fg = MfixM(Mg, g);

    if abs(Fg - MFIin) <= eps
      break;
    elseif Fg > MFIin
      Mb = Mg;
      Fb = Fg;
    else
      Ma = Mg;
      Fa = Fg;
    end
  end

  Mb = Mg;
end

function [Mfi] = MfixM(M, g)
% Computes ht*(mdot/I)^2 for the given M.
  if nargin < 2, g = 1.4; end
  Mfi = g ^ 2 / (g - 1) * M ^ 2 * (1 + (g - 1) / 2 * M ^ 2) / (1 + g * M ^ 2)
^ 2;
end

end
```

Launch Case Iteration

```
clc
close all
clear all


FanPressure = 1.1:0.1:4
phis = [1 2 10]
```

```
SFCs = zeros(length(FanPressure),length(phis));
ISPs = zeros(length(FanPressure),length(phis));
SpecificThrusts = zeros(length(FanPressure),length(phis));
AreaRatios = zeros(length(FanPressure),length(phis));

for i = 1:length(FanPressure)
    for j = 1:length(phis)
        output = AirTurboRocketRamjetIterations(30000,FanPressure(i),phis(j));
        SFCs(i,j) = output(1);
        ISPs(i,j) = output(2);
        SpecificThrusts(i,j) = output(3);
        AreaRatios(i,j) = output(4);
    end
end

plot(FanPressure,SFCs)
title("SFC vs. Fan Pressure Ratio - Launch Case")
xlabel("Fan Pressure Ratio")
ylabel("SFC")
legend('\phi = 1','\phi = 2', '\phi = 10', 'Location', 'NorthEast')

figure
plot(FanPressure,ISPs)
title("Specific Impulse vs. Fan Pressure Ratio - Launch Case")
xlabel("Fan Pressure Ratio")
ylabel("Specific Impulse (s)")
legend('\phi = 1','\phi = 2', '\phi = 10', 'Location', 'NorthWest')

figure
plot(FanPressure,SpecificThrusts)
title("Specific Thrust vs. Fan Pressure Ratio - Launch Case")
xlabel("Fan Pressure Ratio")
ylabel("Specific Thrust")
legend('\phi = 1','\phi = 2', '\phi = 10', 'Location', 'NorthEast')

figure
plot(FanPressure,AreaRatios)
title("Area Ratio vs. Fan Pressure Ratio - Launch Case")
xlabel("Fan Pressure Ratio")
ylabel("Area Ratio")
legend('\phi = 1','\phi = 2', '\phi = 10', 'Location', 'NorthEast')
```

Cruise Case Iteration

```
clc
```

```
close all
clear all


FanPressure = 1.1:0.1:4
phis = [1 2 10]

SFCs = zeros(length(FanPressure),length(phis));
ISPs = zeros(length(FanPressure),length(phis));
SpecificThrusts = zeros(length(FanPressure),length(phis));
AreaRatios = zeros(length(FanPressure),length(phis));

for i = 1:length(FanPressure)
    for j = 1:length(phis)
        output = AirTurboRocketRamjetIterations(60000,FanPressure(i),phis(j));
        SFCs(i,j) = output(1);
        ISPs(i,j) = output(2);
        SpecificThrusts(i,j) = output(3);
        AreaRatios(i,j) = output(4);
    end
end

plot(FanPressure,SFCs)
title("SFC vs. Fan Pressure Ratio - Cruise Case")
xlabel("Fan Pressure Ratio")
ylabel("SFC")
legend('\phi = 1','\phi = 2', '\phi = 10', 'Location', 'NorthEast')

figure
plot(FanPressure,ISPs)
title("Specific Impulse vs. Fan Pressure Ratio - Cruise Case")
xlabel("Fan Pressure Ratio")
ylabel("Specific Impulse (s)")
legend('\phi = 1','\phi = 2', '\phi = 10', 'Location', 'NorthWest')

figure
plot(FanPressure,SpecificThrusts)
title("Specific Thrust vs. Fan Pressure Ratio - Cruise Case")
xlabel("Fan Pressure Ratio")
ylabel("Specific Thrust")
legend('\phi = 1','\phi = 2', '\phi = 10', 'Location', 'NorthEast')

figure
plot(FanPressure,AreaRatios)
title("Area Ratio vs. Fan Pressure Ratio - Cruise Case")
xlabel("Fan Pressure Ratio")
ylabel("Area Ratio")
```

```
legend('\phi = 1','\phi = 2', '\phi = 10', 'Location', 'NorthEast')
```

Air Augumented Rocket Launch Case

Freestream	
Alt (kft)	30
M	0.8
P (Psf)	629.712
Pt (Psf)	959.8951961
T (R)	411.84
Tt (R)	464.55552
q (psf)	281.8866736
Cdo	0.2
A0 (ft^2)	4
U (ft/s)	795.4509745
md (sl/s)	2.834987273
An (ft^2)	0.4
Dn (lbf)	22.55093389
Dram (lbf)	2255.093389

Constants	
gamma	1.4
Mwt_O2	32
Mwt_N2	28.16
Mwt_H2	2
delta_hf	1.30E+09
a	8.80E-05
b	540

Chamber	
Tt1	540
Pt (psf)	200000
md ch	0.1109
phi	10
beta	0
Cp_H2	87010
Cp_O2	5438
md H2	0.06161111111
md O2	0.04928888889
md N2	0
m_brn H2	
Mbp	3.6
R (ft^2/s^2R)	13808.33333
Cp (ft^2/s^2R)	48329.16667
ht_i (ft^2/s^2)	27408120
eta_b	59.60%
Ttb	1884.664272

Throat	
A* (ft^2)	0.0041

Primary Exit	
Mx	4.278246356
Px	915.0778137
Tx	404.3755265
Ux	11961.699
Ae/A*	13.69423813
Ax (ft^2)	0.05657268304
I (lbf)	1342.696329
Htx (ftlb/s)	18035154.12

Inlet	
Pi_d	0.96
Pt (psf)	921.4993883
Tt (R)	464.55552
md	2.834987273
phi	0
beta	3.762
md H2	0
md O2	0.6576124751
md N2	2.177374798
Mb	28.96954389
R (ft^2/s^2R)	1715.94003
Cp (ft^2/s^2R)	6005.790104

Station 15	
M	0.1
Pt (psf)	921.4993883
P (psf)	915.0778137
Tt (R)	464.55552
T (R)	463.6282635
md (sl/s)	2.834987273
md H2	0
md O2	0.6576124751
md N2	2.177374798
Mbp	28.96954389
R (ft^2/s^2R)	1715.94003
Cp (ft^2/s^2R)	6005.790104
A (ft^2)	23.33088503
U (ft/s)	105.6413356
I (lbf)	21649.06711
ht_i (ft^2/s^2)	2790022.945

Station 7	
md (sl/s)	2.945887273
phi	10
md H2	0.06161111111
md O2	0.706901364
md N2	2.177374798
m_brn H2	0
Mbp	25.6601422
R (ft^2/s^2R)	1937.245695
Cp (ft^2/s^2R)	6780.359932
ht_i (ft^2/s^2)	3716788.537
eta_b	0.7492649285
Tt (R)	3510.031182
Ht (ftlb/s)	71512555.88
I (lbf)	22991.76344
A (ft^2)	23.38745771
rho (sl/ft^3)	0.000129081231
U (ft/s)	975.820661
T (R)	3439.811757
M	0.3194819311
P (psf)	860.1664131
Pt (psf)	923.2079258

Nozzle	
Mb	25.6601422
R (ft^2/s^2R)	1937.245695
Cp (ft^2/s^2R)	6780.359932
Tt (R)	3510.031182
T (R)	3146.569927
Pt (psf)	923.2079258
P (psf)	629.712
M	0.7599681119
A (ft^2)	12.84475465
U (ft/s)	2220.089248
md (sl/s)	2.945887273
Cfg	0.96
F (lbf)	6278.527355

Performance	
FN (lbf)	4000.883032
SFCeng	3.210578179
FNWA	1411.252555
ISP (s)	1121.293362
Area Ratio	3.211188662

Air Augumented Rocket Cruise Case

Freestream	
Alt (kft)	60000
M	4
P (Psf)	151.056
Pt (Psf)	22935.62064
T (R)	389.97
Tt (R)	1637.874
q (psf)	1692.559019
Cdo	0.4
A0 (ft^2)	4
U (ft/s)	3870.193587
md (sl/s)	3.498655003
An (ft^2)	0.4
Dn (lbf)	270.8094431
Dram (lbf)	13540.47215

Constants	
gamma	1.4
Mwt_O2	32
Mwt_N2	28.16
Mwt_H2	2
delta_hf	1.30E+09
a	8.80E-05
b	540

Chamber	
Tt1	540
Pt (psf)	200000
md ch	0.0675
phi	10
beta	0
Cp_H2	87010
Cp_O2	5438
md H2	0.0375
md O2	0.03
md N2	0
m_brn H2	
Mbp	3.6
R (ft^2/s^2R)	13808.33333
Cp (ft^2/s^2R)	48329.16667
ht_i (ft^2/s^2)	27408120
eta_b	1
Ttb	1884.664272

Throat	
A* (ft^2)	0.0025

Primary Exit	
Mx	2.344379784
Px	14921.92154
Tx	897.7912258
Ux	9766.755034
Ae/A*	2.283516832
Ax (ft^2)	0.005741764089
I (lbf)	744.0667901
Htx (ftlb/s)	9367582.884

Inlet	
Pi_d	0.669
Pt (psf)	15343.93021
Tt (R)	1637.874
md	3.498655003
phi	0
beta	3.762
md H2	0
md O2	0.8115589081
md N2	2.687096094
Mb	28.96954389
R (ft^2/s^2R)	1715.94003
Cp (ft^2/s^2R)	6005.790104

Station 15	
M	0.2
Pt (psf)	15343.93021
P (psf)	14921.92154
Tt (R)	1637.874
T (R)	1624.875
md (sl/s)	3.498655003
md H2	0
md O2	0.8115589081
md N2	2.687096094
Mbp	28.96954389
R (ft^2/s^2R)	1715.94003
Cp (ft^2/s^2R)	6005.790104
A (ft^2)	1.647832708
U (ft/s)	396.7211103
I (lbf)	25976.82068
ht_i (ft^2/s^2)	9836727.46

Station 7	
md (sl/s)	3.566155003
phi	10
md H2	0.0375
md O2	0.8415589081
md N2	2.687096094
m_brn H2	0
Mbp	27.20540125
R (ft^2/s^2R)	1827.21069
Cp (ft^2/s^2R)	6395.237416
ht_i (ft^2/s^2)	10169317.88
eta_b	0.7857930415
Tt (R)	3223.0089
Ht (ftlb/s)	74864739.66
I (lbf)	26720.88747
A (ft^2)	1.653574472
rho (sl/ft^3)	0.002469872392
U (ft/s)	873.1763033
T (R)	3163.399162
M	0.306949275
P (psf)	14276.34841
Pt (psf)	15240.29669

Nozzle	
Mb	27.20540125
R (ft^2/s^2R)	1827.21069
Cp (ft^2/s^2R)	6395.237416
Tt (R)	3223.0089
T (R)	862.4449403
Pt (psf)	15240.29669
P (psf)	151.056
M	3.699365133
A (ft^2)	6.770660253
U (ft/s)	5494.791526
md (sl/s)	3.566155003
Cfg	0.96
F (lbf)	18811.46716

Performance	
FN (lbf)	5000.185562
SFCeng	1.563598371
FNWA	1429.173656
ISP (s)	2302.381524
Area Ratios	1.692665063

Air Turbo Ramjet Rocket Launch Case

Freestream	
Alt (kft)	30000
M	0.8
P (Psf)	629.712
Pt (Psf)	959.8951961
T (R)	411.84
Tt (R)	464.55552
q (psf)	281.8866736
Cdo	0.2
A0 (ft^2)	2
U (ft/s)	795.4509745
md (sl/s)	1.417493637
An (ft^2)	0.2
Dn (lbf)	11.27546694
Dram (lbf)	1127.546694

Constants	
gamma	1.4
Mwt_O2	32
Mwt_N2	28.16
Mwt_H2	2
delta_hf	1.30E+09
a	8.80E-05
b	540

Fan	
Pi_f	2
eta_f	0.95
Pt (psf)	1842.998777
Tt (R)	571.6544695
md	4.507629764
Pwr (ft-lbf/s)	2899369.719
md H2	0
md O2	1.045603835
md N2	3.462025929
Mb	28.96954389
R (ft^2/s^2R)	1715.94003
Cp (ft^2/s^2R)	6005.790104

Station 15	
M	0.2
Pt (psf)	1842.998777
P (psf)	1792.310234
Tt (R)	571.6544695
T (R)	567.1175293
md (sl/s)	4.507629764
A (ft^2)	10.44235281
U (ft/s)	234.3753146
I (lbf)	19772.41294
Ht (ftlb/s)	15475760.19
md H2	0
md O2	1.045603835
md N2	3.462025929
Mbp	28.96954389
R (ft^2/s^2R)	1715.94003
Cp (ft^2/s^2R)	6005.790104

Dummy Calc for Ttb	
Tt1	540
Pt (psf)	200000
md ch	1
phi	2
beta	0
Cp_H2	87010
Cp_O2	5438
md H2	0.2
md O2	0.8
md N2	0
m_brn H2	
Mbp	10
R (ft^2/s^2R)	4971
Cp (ft^2/s^2R)	17398.5
ht_i (ft^2/s^2)	11746296
eta_b	0.5961337334
Ttb	5129.389393

Turbine Exit	
eta_t	0.9
eta_mu	0.99
Mx	1
Px	1792.310234
Ptx	3392.716303
Tx	1627.707655
Ttx	1953.249186
dh	13847524
md	0.05299768886
phi	2
beta	0
md H2	0.01059953777
md O2	0.04239815108
md N2	0
Mbp	10
R (ft^2/s^2R)	4971
Cp (ft^2/s^2R)	17398.5
Ux (ft/s)	3365.689922
Ax (ft^2)	0.07108693889
I (lbf)	261.0193369

Station 7	
md (sl/s)	4.560627453
md H2	0.01059953777
md O2	1.088001987
md N2	3.462025929
Mbp	28.58006111
R (ft^2/s^2R)	1739.324483
Cp (ft^2/s^2R)	6087.635689
ht_i (ft^2/s^2)	2894101.118
eta_b	0.7887136307
Tt (R)	953.6543613
Ht (ftlb/s)	26860955.85
I (lbf)	20033.43228
A (ft^2)	10.51343975
rho (sl/ft^3)	0.00105677001
U (ft/s)	410.4868861
T (R)	939.8148765
M	0.2713462391
P (psf)	1727.441725
Pt (psf)	1818.125254

Nozzle	
Mb	28.58006111
R (ft^2/s^2R)	1739.324483
Cp (ft^2/s^2R)	6087.635689
Tt (R)	953.6543613
T (R)	704.4087826
Pt (psf)	1818.125254
P (psf)	629.712
M	1.330106307
A (ft^2)	5.093718534
U (ft/s)	1742.019679
md (sl/s)	4.560627453
Cfg	0.96
F (lbf)	7626.914659

Performance	
FN (lbf)	4005.460185
SFCeng	1.532540888
ISP (s)	2349.0401
FNWA	888.5956467
FN/Amx	628.9308176
Area Ratios	0.8008991406

Air Turbo Ramjet Rocket Cruise Case

Freestream	
Alt (kft)	30
M	0.8
P (Psf)	629.712
Pt (Psf)	959.8951961
T (R)	411.84
Tt (R)	464.55552
q (psf)	281.8866736
Cdo	0.2
A0 (ft^2)	4
U (ft/s)	795.4509745
md (sl/s)	2.834987273
An (ft^2)	0.4
Dn (lbf)	22.55093389
Dram (lbf)	2255.093389

Constants	
gamma	1.4
Mwt_O2	32
Mwt_N2	28.16
Mwt_H2	2
delta_hf	1.30E+09
a	8.80E-05
b	540

Fan	
Pi_f	2
eta_f	0.95
Pt (psf)	30687.86042
Tt (R)	2015.4706
md	1.845540514
Pwr (ft-lbf/s)	4185253.886
md H2	0
md O2	0.428097324
md N2	1.41744319
Mb	28.96954389
R (ft^2/s^2R)	1715.94003
Cp (ft^2/s^2R)	6005.790104

Station 15	
M	0.2
Pt (psf)	30687.86042
P (psf)	29843.84308
Tt (R)	2015.4706
T (R)	1999.474801
md (sl/s)	1.845540514
A (ft^2)	0.4821182556
U (ft/s)	440.0816903
I (lbf)	15200.45015
Ht (ftlb/s)	22339332.94
md H2	0
md O2	0.428097324
md N2	1.41744319
Mbp	28.96954389
R (ft^2/s^2R)	1715.94003
Cp (ft^2/s^2R)	6005.790104

Dummy Calc for Ttb	
Tt1	540
Pt (psf)	200000
md ch	1
phi	2
beta	0
Cp_H2	87010
Cp_O2	5438
md H2	0.2
md O2	0.8
md N2	0
m_brn H2	
Mbp	10
R (ft^2/s^2R)	4971
Cp (ft^2/s^2R)	17398.5
ht_i (ft^2/s^2)	11746296
eta_b	0.5961337334
Ttb	5129.389393

Turbine Exit	
eta_t	0.9
eta_mu	0.99
Mx	1
Px	29843.84308
Ptx	56492.28077
Tx	3108.210186
Ttx	3729.852223
dh	24890690.69
md	0.173616249
phi	2
beta	0
md H2	0.0347232498
md O2	0.1388929992
md N2	0
Mbp	10
R (ft^2/s^2R)	4971
Cp (ft^2/s^2R)	17398.5
Ux (ft/s)	4650.943771
Ax (ft^2)	0.01932629812
I (lbf)	1381.331067

Station 7	
md (sl/s)	2.019156763
md H2	0.0347232498
md O2	0.5669903232
md N2	1.41744319
Mbp	26.31596277
R (ft^2/s^2R)	1888.96756
Cp (ft^2/s^2R)	6611.386461
ht_i (ft^2/s^2)	10000920.82
eta_b	0.6509894717
Tt (R)	3895.602632
Ht (ftlb/s)	53014748.78
I (lbf)	16581.78122
A (ft^2)	0.5014445538
rho (sl/ft^3)	0.004024463113
U (ft/s)	1000.550856
T (R)	3819.892187
M	0.3148019019
P (psf)	29039.12703
Pt (psf)	31103.98

Nozzle	
Mb	26.31596277
R (ft^2/s^2R)	1888.96756
Cp (ft^2/s^2R)	6611.386461
Tt (R)	3895.602632
T (R)	850.2052806
Pt (psf)	31103.98
P (psf)	151.056
M	4.231994416
A (ft^2)	3.382961869
U (ft/s)	6345.7543
md (sl/s)	2.019156763
Cfg	0.96
F (lbf)	12300.5498

Performance	
FN (lbf)	5015.098759
SFCeng	4.009760539
ISP (s)	897.8092246
FNWA	2717.414612
FN/Amx	2369.668246
Area Ratios	1.603299464